IONOS Managed Kubernetes Deployment & Flow Guide

Service: privacy-detection-service | **Cluster:** IONOS Managed K8s | **DB:** 87.106.50.234:5432/hekmaai

1. IONOS Kubernetes Production Topology & Traffic Flow

The microservice deploys as a highly available, auto-scaling Kubernetes workload on IONOS Cloud, routing HTTPS 443 traffic to container port 8080 and storing token mappings directly in the existing application database (hekmaai).

2. Step-by-Step Deployment Runbook (Commands)

```
# 1. Apply Namespace, ConfigMap & Secret:
kubectl apply -f k8s/namespace.yaml
kubectl apply -f k8s/configmap.yaml
kubectl apply -f k8s/secret.yaml

# 2. Run Database Migration Job (Initializes 4 vault tables in hekmaai):
kubectl apply -f k8s/migration-job.yaml
kubectl logs -n privacy-detection job/privacy-detection-db-migration -f

# 3. Deploy Workload, Service & Autoscaler:
kubectl apply -f k8s/deployment.yaml
kubectl apply -f k8s/service.yaml
kubectl apply -f k8s/hpa.yaml

# 4. Apply Ingress for Custom Domain (pii-phi.hekma.ai):
kubectl apply -f k8s/ingress.yaml
kubectl rollout status deployment/privacy-detection-service -n privacy-detection
```

3. Kubernetes Manifests Summary in Repository (k8s/)

Manifest File	Resource Type	Configuration Details
k8s/namespace.yaml	Namespace	privacy-detection (Dedicated tenant isolation)
k8s/configmap.yaml	ConfigMap	CUSTOM_DOMAIN=https://pii-phi.hekma.ai, STORAGE=postgresql
k8s/secret.yaml	Secret	DATABASE_URL (87.106.50.234:5432/hekmaai), OPENAI_API_KEY
k8s/deployment.yaml	Deployment	2 Replicas, Requests: 1 CPU/2Gi, Limits: 2 CPU/4Gi, /ready probe
k8s/service.yaml	Service	ClusterIP exposing Port 8080
k8s/ingress.yaml	Ingress	NGINX Ingress routing pii-phi.hekma.ai (TLS Enabled)
k8s/hpa.yaml	Autoscaler	Auto-scales 2 to 6 pods at 70% CPU / 80% RAM
k8s/migration-job.yaml	Job	Creates token_mappings in hekmaai before pod traffic

4. Observability Probes & Instant Rollback

- **Liveness Probe:** GET /health on port 8080 (initial delay: 15s)
- **Readiness Probe:** GET /ready on port 8080 (initial delay: 20s, ensures RoBERTa model is warm in RAM)
- **Rollback Command:** kubectl rollout undo deployment/privacy-detection-service -n privacy-detection
- **Live Pod Logs:** kubectl logs -n privacy-detection -l app=privacy-detection-service -f --tail=100